CSCI 1101 Introduction to Computer Science

Data Representation

Overview

Definitions

- Data
- Data Representation

Number Systems

- Binary
- Decimal
- Hexadecimal

Conversions

- From Decimal
- From Binary
- From Hexadecimal

Computers & Data

- Representing Numbers
- Representing Text
- Representing Images
- Representing Sound
- Representing Video

Definitions

What is Data?

- Think back to projects, experiments, and other previous experiences.

Definition:

Characteristics or attributes that represent people, events, things, and ideas

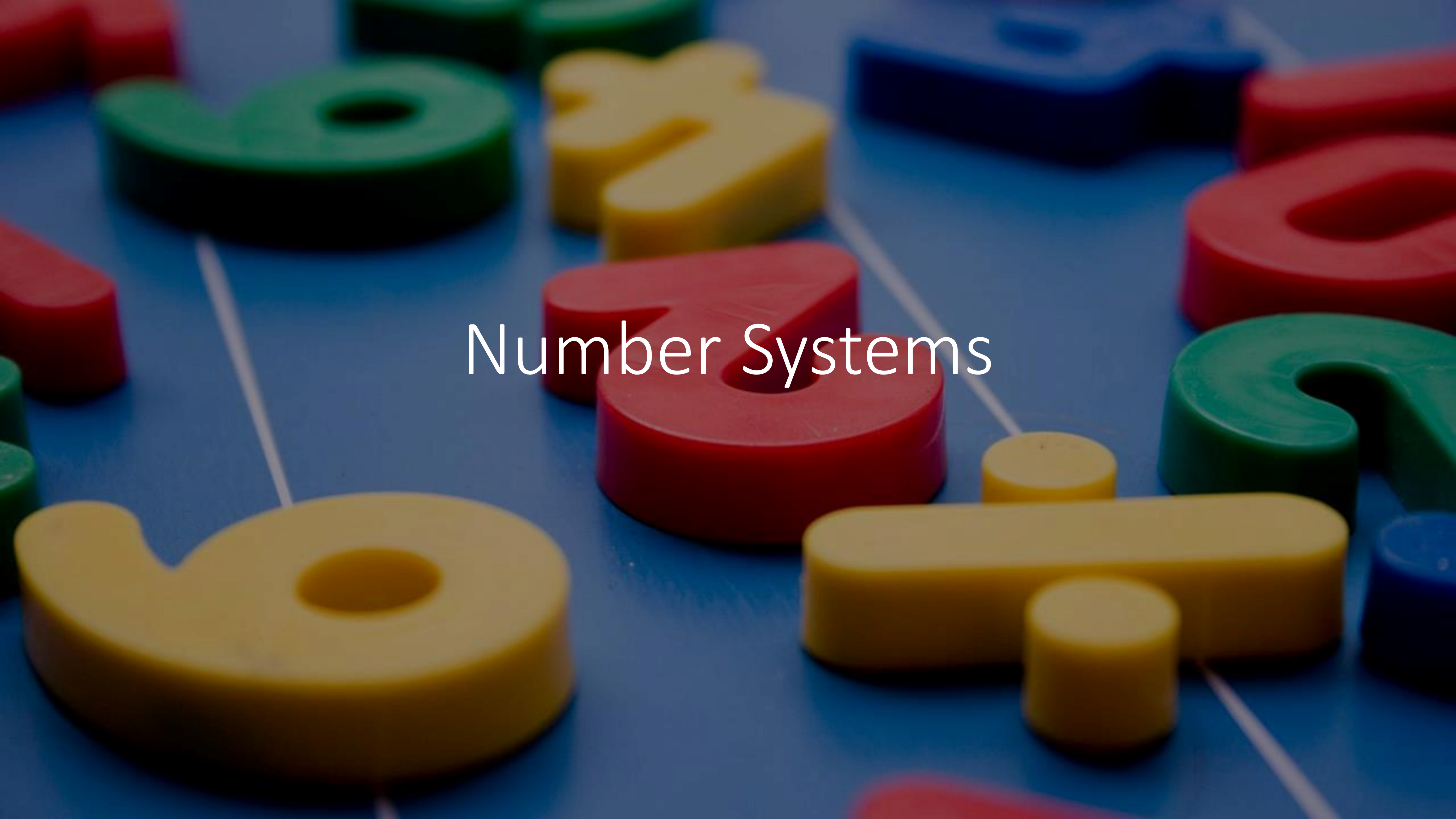
Thus, data can be a:

- Name
- Number (measurement)
- Color
- Etc.

Computers and Data

- With computers, smartphones, and other devices, one can listen to music, watch videos, play games, and more. The music, videos, games, and other information are the computer’s data.
- Data Representation refers to the form in which data is stored, processed, and transmitted.
- Computers and other devices store data in digital (data expressed in Binary) formats that can be handled by electronic circuitry.

Number Systems



Decimal Numbers

- Think about the numbers, you learned at school while growing up.
- They had 10 digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
- The number 1,248 is

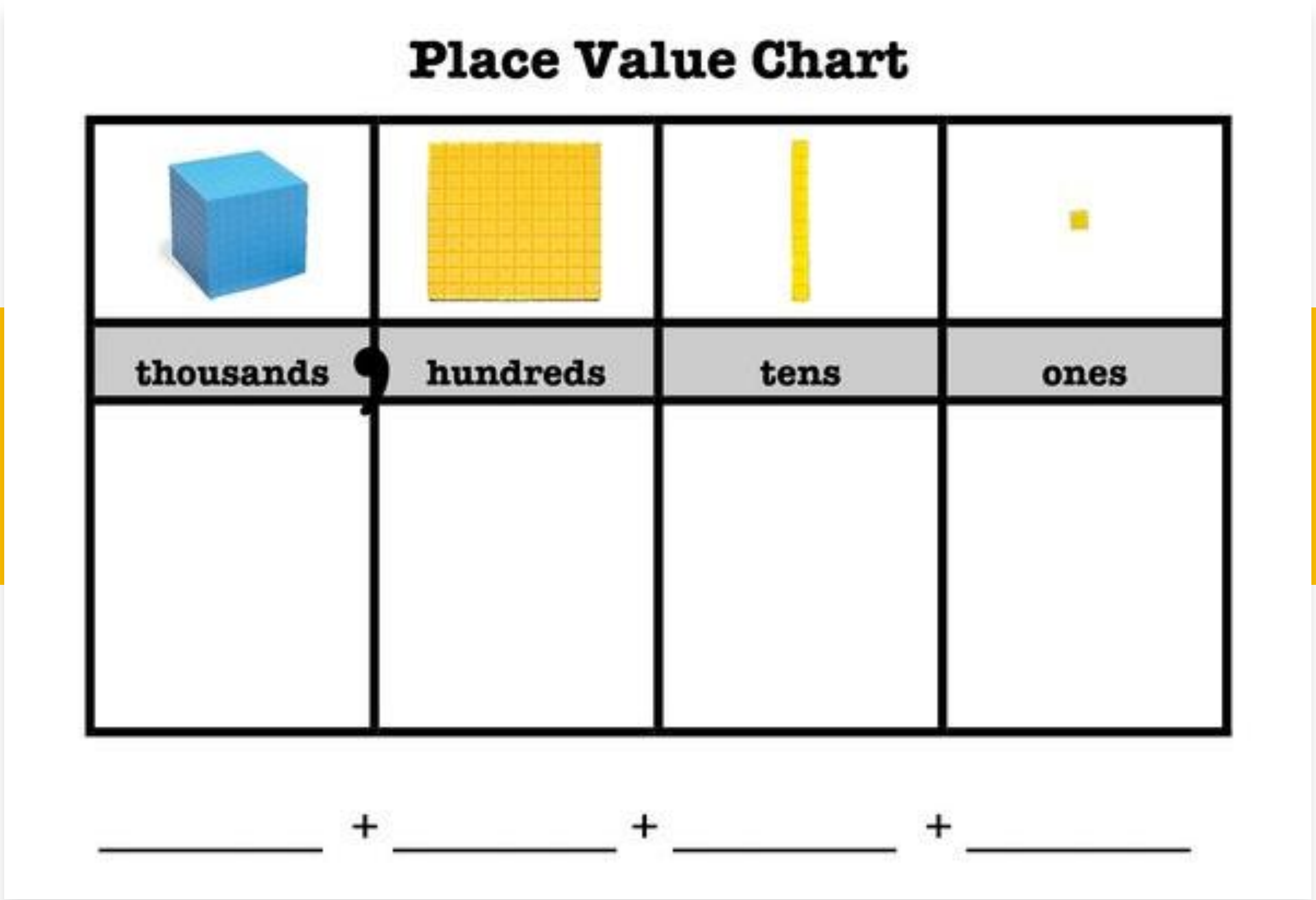
$$1{,}000 + 200 + 40 + 8$$

- We can change the representation to:

$$(1 * 1{,}000) + (2 * 100) + (4 * 10) + (8 * 1)$$

- We can change it even further to:

$$(1 * 10^3) + (2 * 10^2) + (4 * 10^1) + (8 * 10^0)$$



Decimal Numbers (cont.)

Binary

- Digits: 0, 1
- Base 2
- Binary digit → *bit*

2^3	2^2	2^1	2^0

______________ + ______________ + ______________ + ______________

Hexadecimal

- Digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F
- Base 16

16^3	16^2	16^1	16^0

______________ + ______________ + ______________ + ______________

Conversions

Decimal (Base 10)	Binary (Base 2)	Hexadecimal (Base 16)
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Converting From Decimal

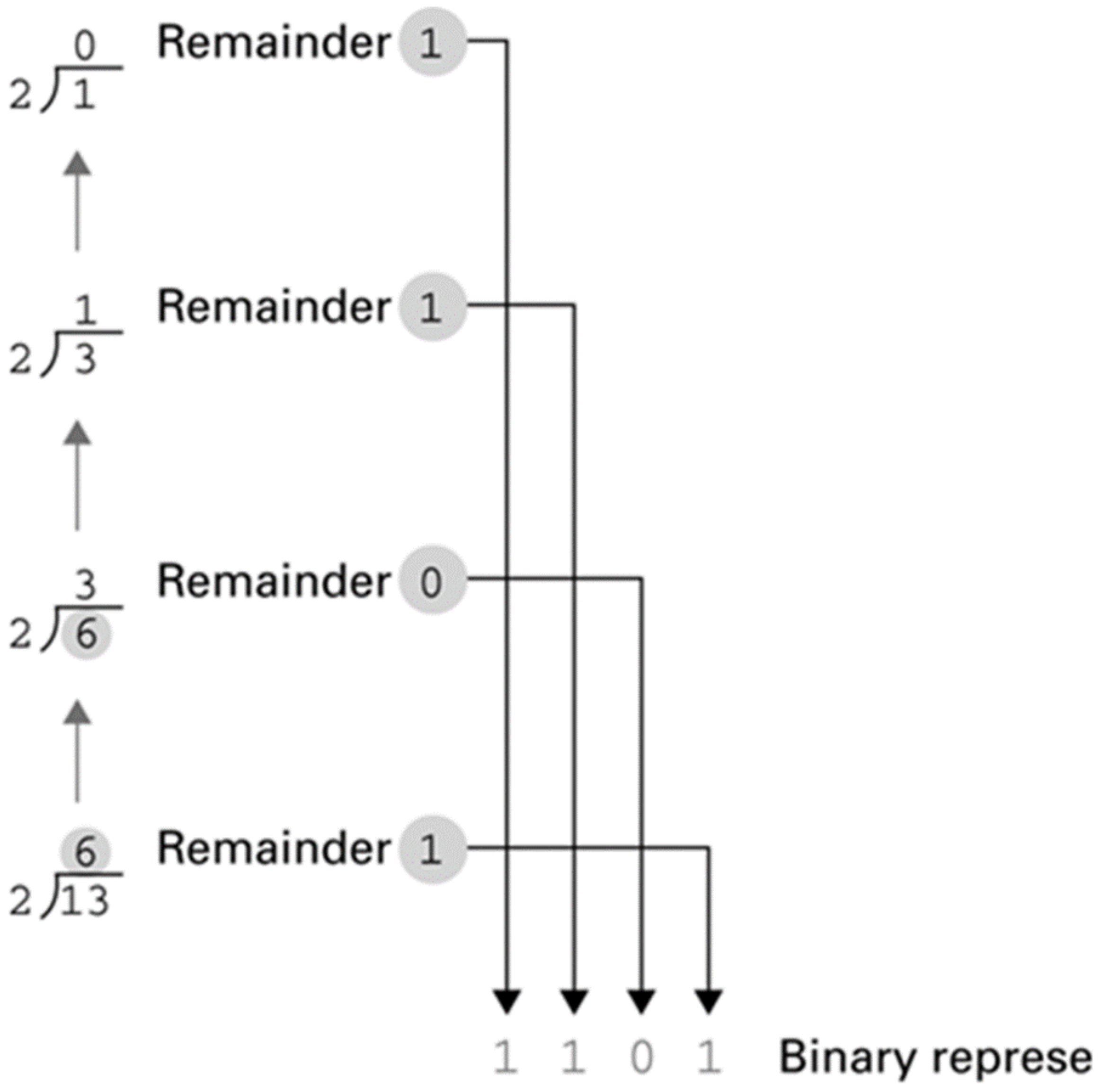
Step 1. Divide the value by the desired base number (B) and record the remainder.

Step 2. As long as the quotient obtained is not zero, continue to divide the newest quotient by B and record the remainder.

Step 3. Now that a quotient of zero has been obtained, the new number system representation of the original value consists of the remainders listed from right to left in the order they were recorded.

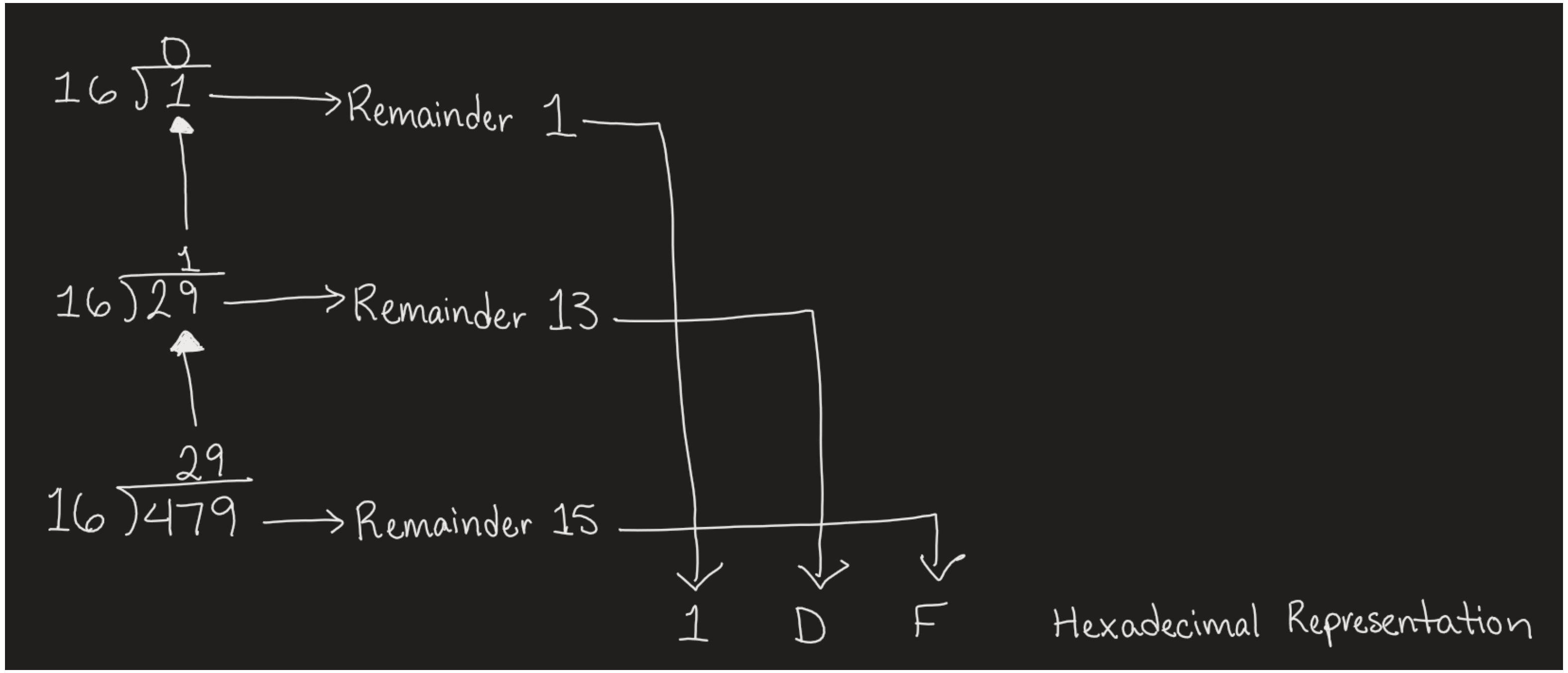
Example 1-1

Start here →



Example 1-2

Start here →



Converting To Decimal

Step 1.

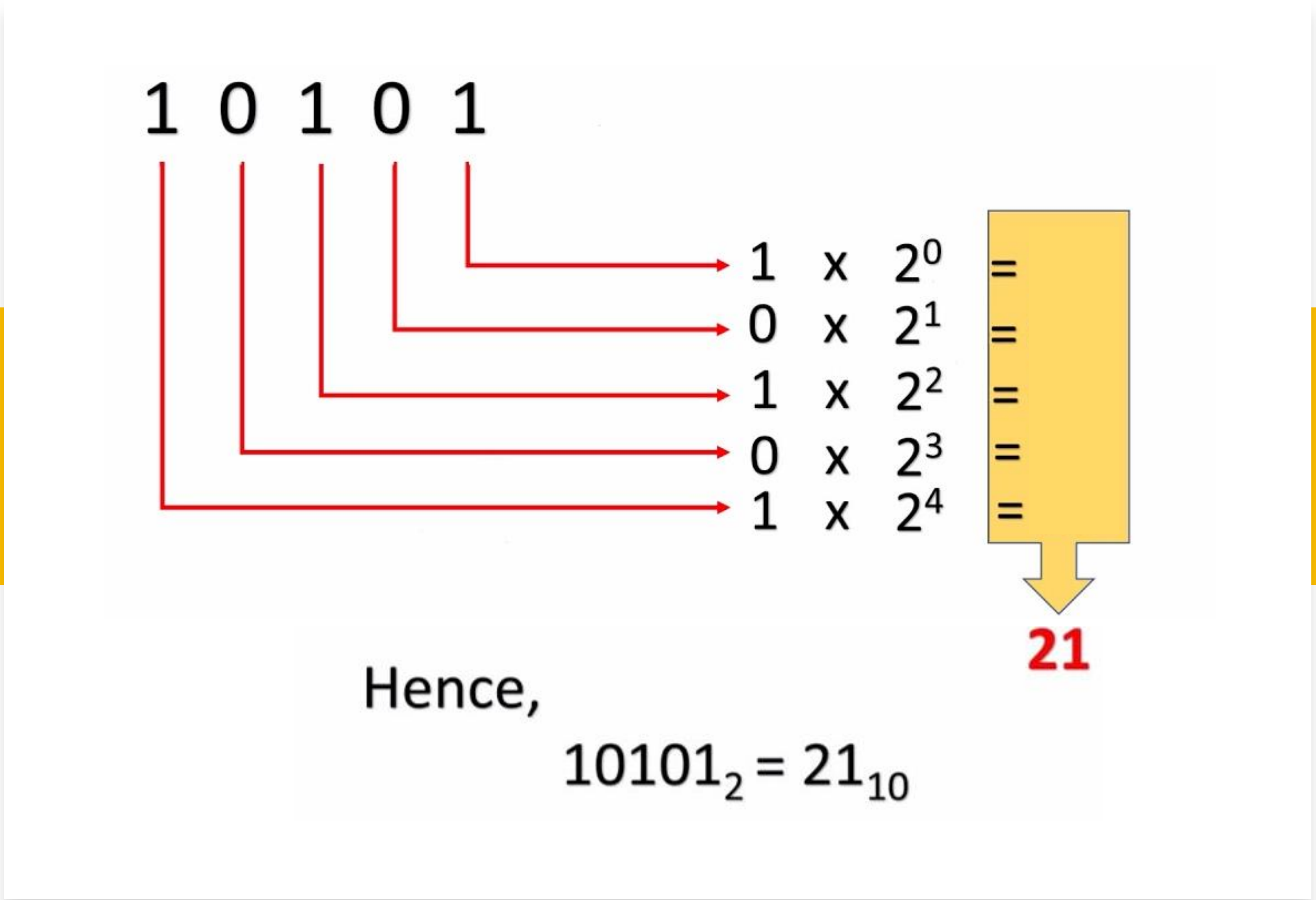
Multiply the digit value to the quantity it represents (B^n) and record the value. (Recommended to start frt

Step 2.

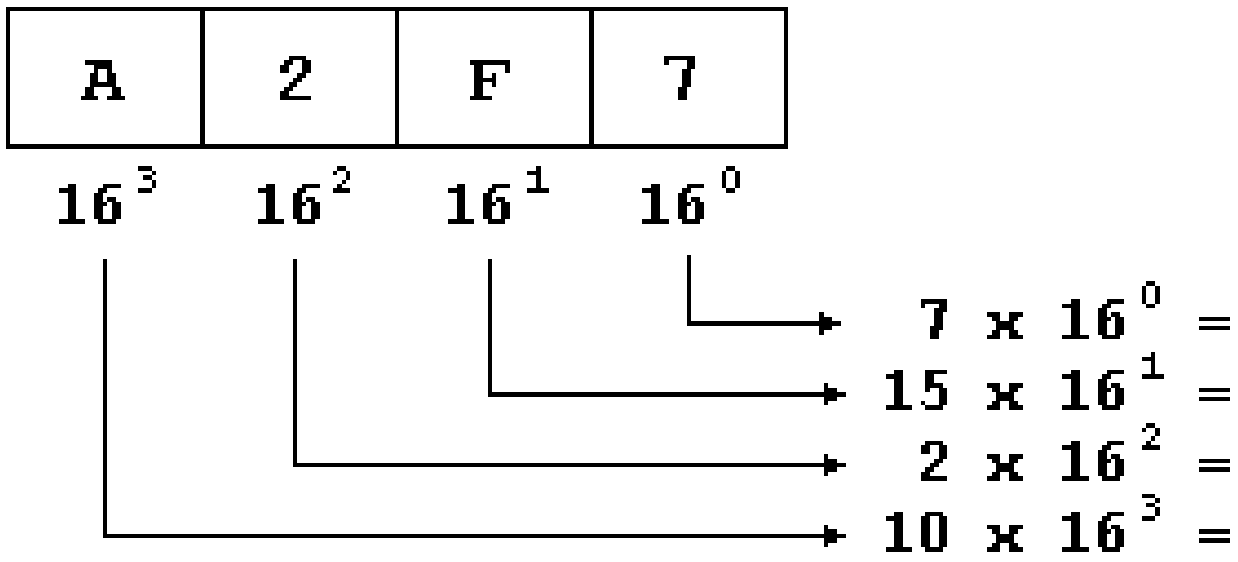
Repeat step 1 until no digits in the number given are left.

Step 3.

Add all the recorded values together to obtain the equivalent decimal



Example 2-1



Example 2-2

Converting From Binary To Hexadecimal

Step 1. Mark groups of four bits (starting at the right end)

Step 2. Convert each group of bits into hex digits.

Bit pattern	Hexadecimal representation
0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7
1000	8
1001	9
1010	A
1011	B
1100	C
1101	D
1110	E
1111	F

10101011

$\underline{1010}$ $\underline{1011}$

A B

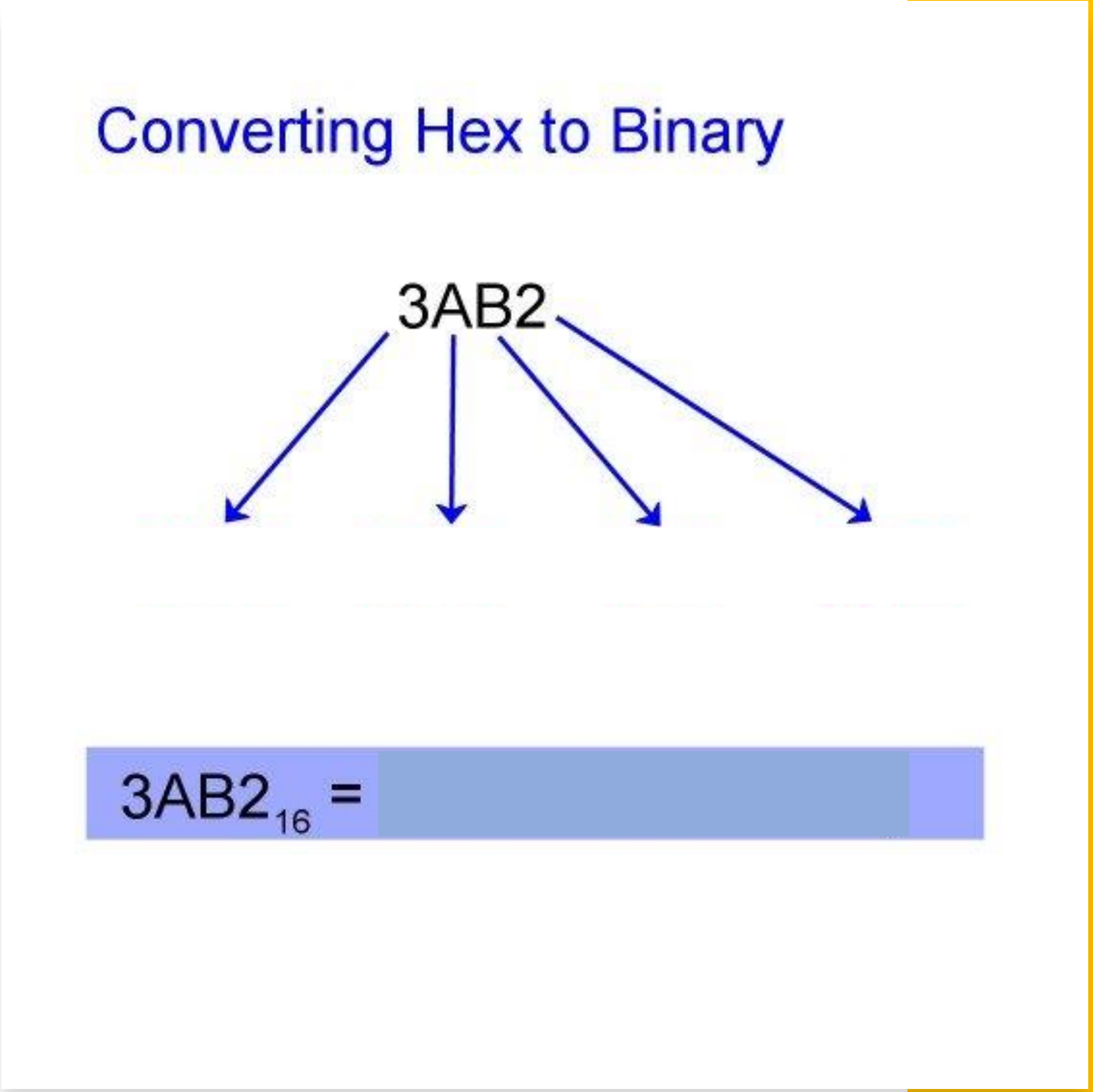
Example 3

Converting From Hexadecimal To Binary

Step 1. Convert each hex digit into binary from left to right

Bit pattern	Hexadecimal representation
0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7
1000	8
1001	9
1010	A
1011	B
1100	C
1101	D
1110	E
1111	F

Example 4



Converting From Binary To Hexadecimal

Example 5

Binary --> 101101

Decimal

Hexadecimal

Converting From Binary To Hexadecimal

Example 6

Binary --> 011000110

Decimal

Hexadecimal

Converting From Hexadecimal To Binary

Example 7

Hexadecimal --> 54

Binary

Decimal

Converting From Hexadecimal To Binary

Example 8

Hexadecimal --> 25B

Binary

Decimal

Converting From Decimal To Binary

Example 9

Decimal --> 111

Division	**Quotient**	**Remainder**

Converting From Decimal To Binary

Example 10

Decimal --> 255

Division	**Quotient**	**Remainder**

Converting From Decimal To Hexadecimal

Example 11

Decimal --> 204

Division	**Quotient**	**Remainder**

Converting From Decimal To Hexadecimal

Example 12

Decimal --> 2898

Division	**Quotient**	**Remainder**

Arithmetic

Arithmetic in Binary: Adding Binary Numbers

Remember that there are only 2 digits in binary, 0 and 1

Carry Values

			1	1	1	1	1	
		1	0	1	0	1	1	1
+		1	0	0	1	0	1	1
	1	0	1	0	0	0	1	0*

*****: 1+1 = 2
No digit for 2 in Binary. So 2-2=**0** and a **carry**

Subtracting Binary Numbers

Remember borrowing? Apply that concept here:

	$\cancel{1}^{0}$	$0^{\cancel{2}\ 1}$	$\cancel{1}^{0\ 2}$	0^{2}	1	1	1
-		1	1	1	0	1	1
	0	0	1	1	1	0	0

If minuend < subtrahend borrow **2** from immediate left column and subtract **1** from value in that column.

Same idea, different base (hex)

- Addition

$$
\begin{array}{rcc}
 & 1 & \\
 & A & 3 \\
+ & 1 & E \\
\hline
 & C & 1^{*}
\end{array}
$$

*: 3+E = 3+14=17
No digit for 17 in Hex
So 17-16=**1** and a **carry**

- Subtraction

$$
\begin{array}{rcc}
 & \cancel{A}^{9} & 3^{16} \\
- & 1 & E \\
\hline
 & 8 & 5
\end{array}
$$

If minuend < subtrahend borrow **16** from immediate left column and subtract **1** from value in that column.

Example 13

	1	0	1	1
+	0	1	1	1

Example 14

$$
\begin{array}{rrrrrrr}
 & 1 & 0 & 1 & 0 & 1 & 0 \\
+ & & 1 & 1 & 0 & 1 & 1 \\
\hline
\end{array}
$$

Example 15

	1	0	1	1
-	0	1	1	1

Example 16

$$\begin{array}{r} 1\ 0\ 1\ 0\ 1\ 0 \\ -\quad 1\ 1\ 0\ 1\ 1 \\ \hline \end{array}$$

Example 17

$$\begin{array}{rrr} & 9 & 2 \\ + & 4 & 8 \\ \hline \end{array}$$

Example 18

	F	9	2
+	E	4	8

Example 19

$$\begin{array}{rr} & 9\ \ 2 \\ - & 4\ \ 8 \\ \hline \end{array}$$

Example 20

	F	A	1
-	E	B	A

Representing Numbers

Representing Numeric Values

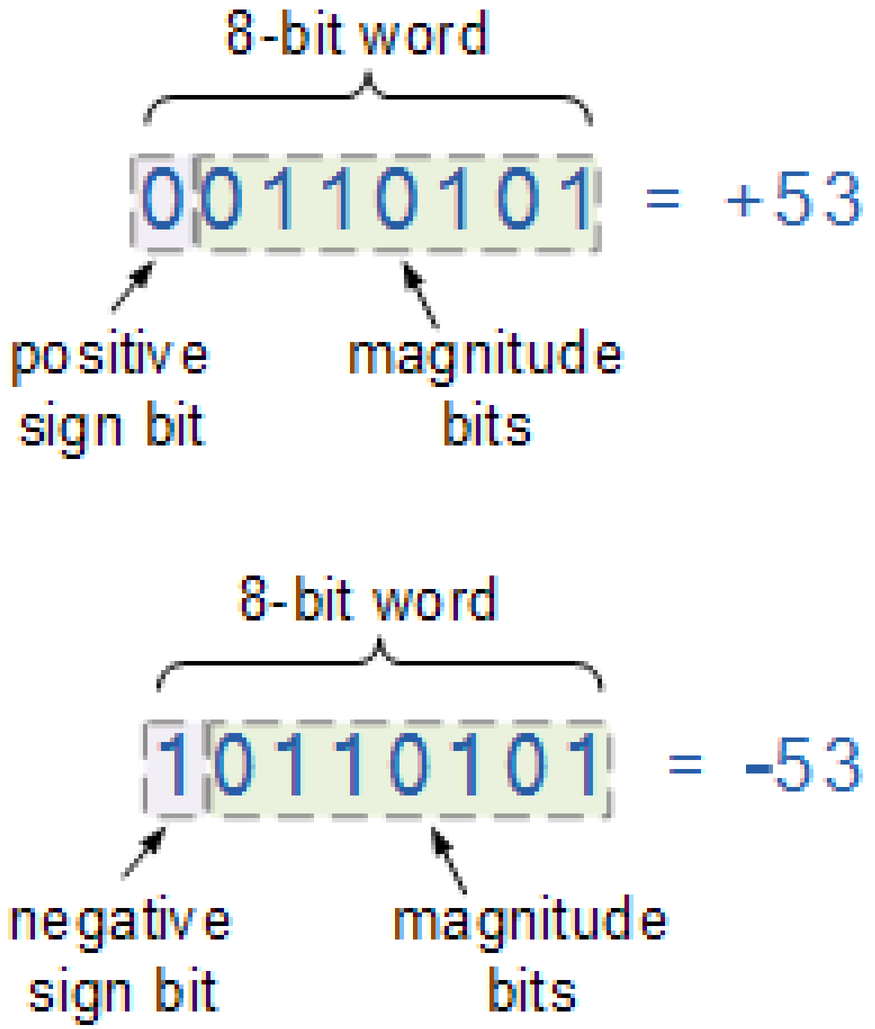
- Using the number system (binary, hexadecimal), whole numbers (0, 1, 2,…) can be represented
- What about integers and floating-point (real) numbers?
  - Modify binary system to represent these values as well

Representing Negative Values

Signed-Magnitude (SM) Notation

1’s Complement

2’s Complement



Sign-Magnitude (SM) Notation

One bit of the binary number is used to determine the sign as positive (0) or negative (1). The rest of the binary digits are used to represent the magnitude, or value, of the binary number in the usual way.

What is the problem with SM Notation?

There are two representations for 0, a positive and negative representation.

1’s Complement

Positive numbers remain unchanged and start with a 0. Negative numbers are obtained by inverting the unsigned positive number, or changing the 0 to 1 and 1 to 0. Since the positive number always starts with 0, the complements will always start with a 1 to indicate negative numbers.

Example: 8-bit representation of $\pm$12

+12: 0000 1100

-12: 1111 0011

What is the problem with 1’s complement?

It also has two representations for 0.

+0: 0000

- 0: 1111

2’s Complement

Positive numbers remain unchanged. Negative numbers are obtained by adding 1 to the least significant bit of the 1’s complement of the number. Since the positive number always starts with 0, the complements will always start with a 1 to indicate negative numbers.

Example: 8-bit representation of $\pm$12

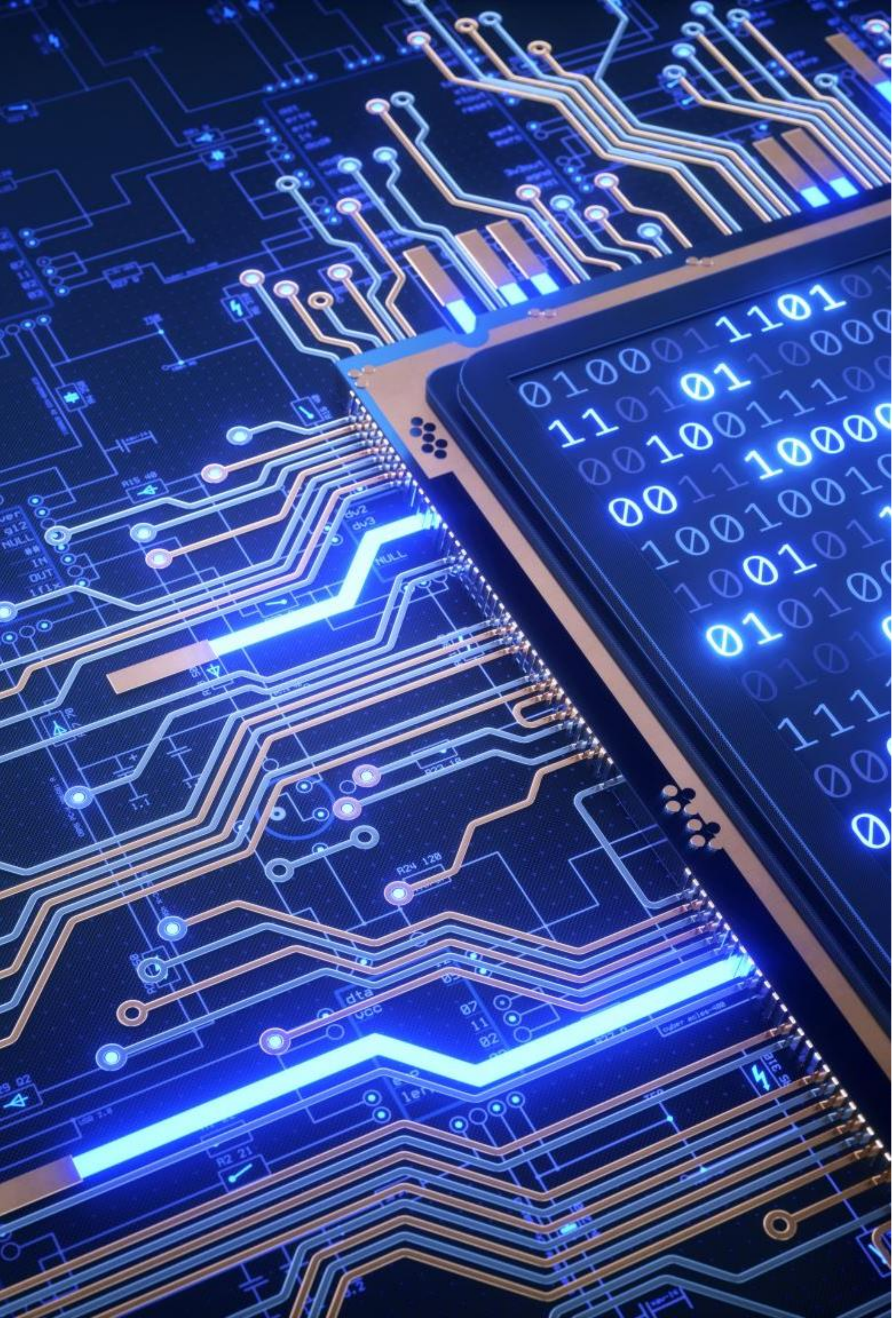
+12: 0000 1100

-12: 1111 0100

Does 2’s Complement have two representations for 0?

> No, it does not, making this notation more widely used than the other two.

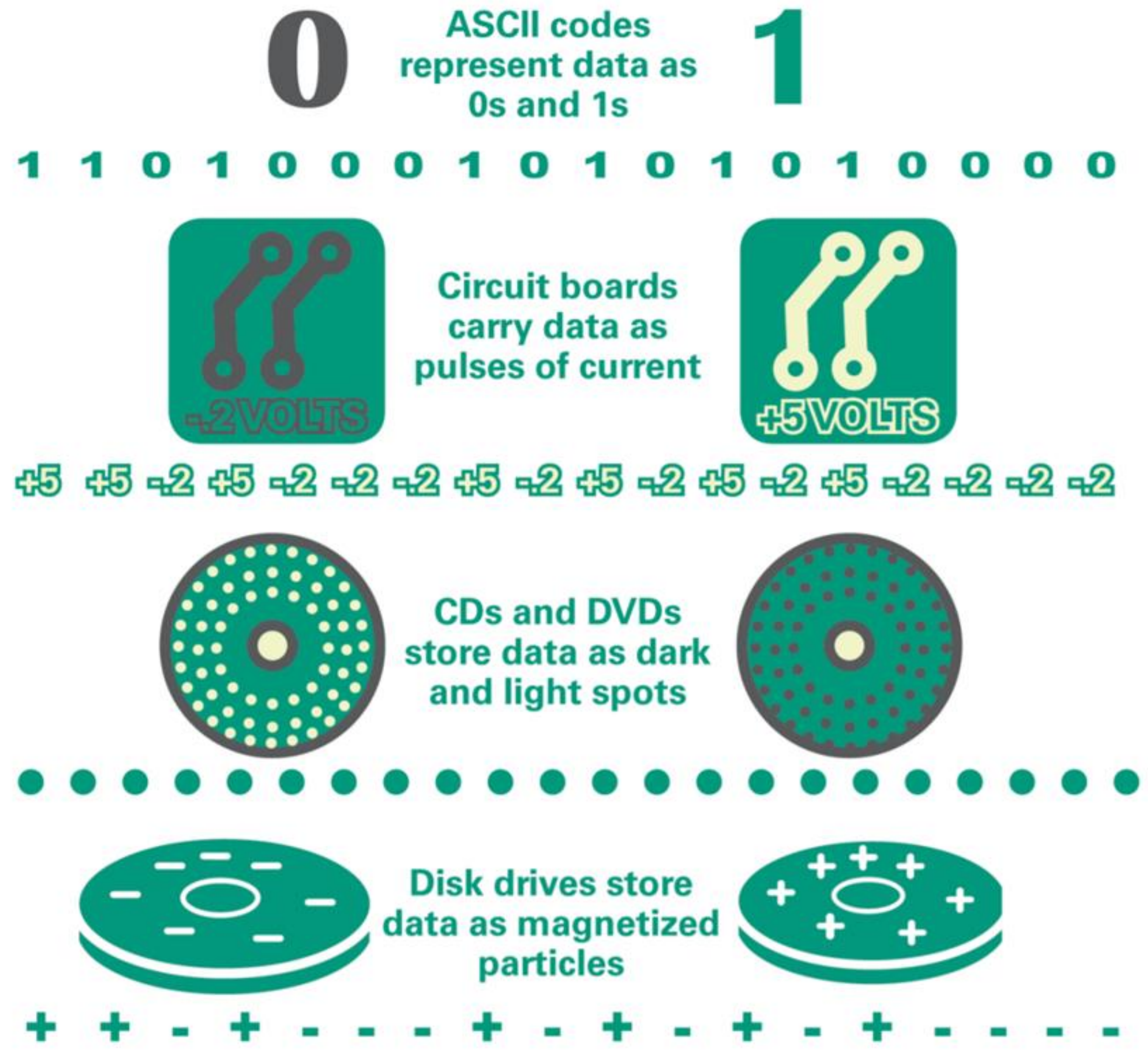
Computers and Data



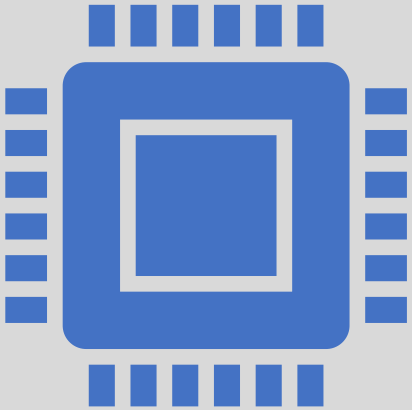
Why Binary?

- Computers operate with voltage through what amounts to microscopic switches that are either “on” (1) or “off” (0).

Low Voltage = 0
High Voltage = 1


0
ASCII codes represent data as 0s and 1s
1
1 1 0 1 0 0 0 1 0 1 0 1 0 1 0 0 0 0
-2 VOLTS
Circuit boards carry data as pulses of current
+5 VOLTS
+5 +5 -2 +5 -2 -2 -2 +5 -2 +5 -2 +5 -2 +5 -2 -2 -2 -2
CDs and DVDs store data as dark and light spots
Disk drives store data as magnetized particles
+ + - + - - - + - + - + - + - - - -

Bits and Bytes



- Terminology related to bits and bytes is extensively used to describe storage capacity and network access speed.
  - Bits → binary digit
  - Byte → group of eight bits (abbreviated as an uppercase B)
- Have you ever heard phrases like “megabytes”, “terabytes”, or something similar?

Unit	Abbreviation	Approximate Size
Bit	b	Binary digit, single 1 or 0
Nibble	–	4 bits
Byte	B	8 bits
Kilobyte	KB	1,024 bytes or 10^3
Megabyte	MB	1,024 KB or 10^6
Gigabyte	GB	1,024 MB or 10^9
Terabyte	TB	1,024 GB or 10^{12}
Petabyte	PB	1,024 TB or 10^{15}
Exabyte	EB	1,024 PB or 10^{18}
Zettabyte	ZB	1,024 EB or or 10^{21}

Representing Text

ASCII	Unicode
7 bits per character	16, 24, or 32 bits per character
128 different characters	Between 65,536 – 4,294,967,296 characters depending on the system ASCII
All English letters, numbers, and symbols	Contains all major languages, symbols, hieroglyphics, etc.

ASCII Code

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[ENG OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	|
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]

ASCII Code

Table 1.2 The seven-bit ASCII code.

Bit positions	**Bit positions 654**							
3210	000	001	010	011	100	101	110	111
0000	NUL	DLE	SPACE	0	@	P	`	p
0001	SOH	DC1	!	1	A	Q	a	q
0010	STX	DC2	"	2	B	R	b	r
0011	ETX	DC3	#	3	C	S	c	s
0100	EOT	DC4	$	4	D	T	d	t
0101	ENQ	NAK	%	5	E	U	e	u
0110	ACK	SYN	&	6	F	V	f	v
0111	BEL	ETB	'	7	G	W	g	w
1000	BS	CAN	(	8	H	X	h	x
1001	HT	EM	)	9	I	Y	i	y
1010	LF	SUB	*	:	J	Z	j	z
1011	VT	ESC	+	;	K	[	k	{
1100	FF	FS	,	<	L	\	l	|
1101	CR	GS	-	=	M	]	m	}
1110	SO	RS	.	>	N	^	n	~
1111	SI	US	/	?	O	—	o	DEL

Example:

Tom 2_3 →

0101 0100
0110 1111
0110 1101
0010 0000
0011 0010
0101 1111
0011 0011

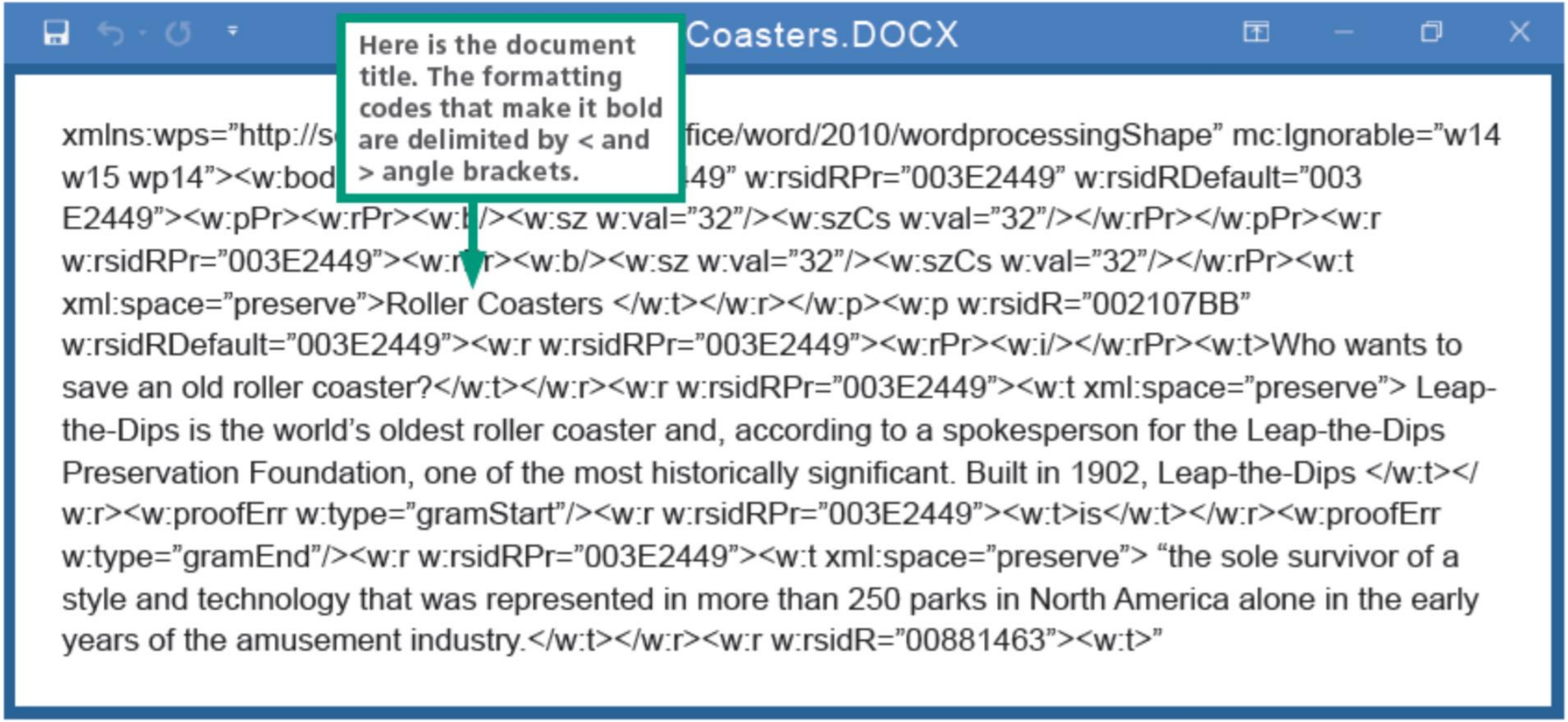
https://www.ascii-code.com/

ASCII Applications

- Used for numerals
  - Social Security numbers
  - Phone numbers
- Plain, unformatted text (ASCII text) stored in text file (.txt)
  - Apple: “Plain Text”
  - Windows: “Text Document”

Formatted Text Files

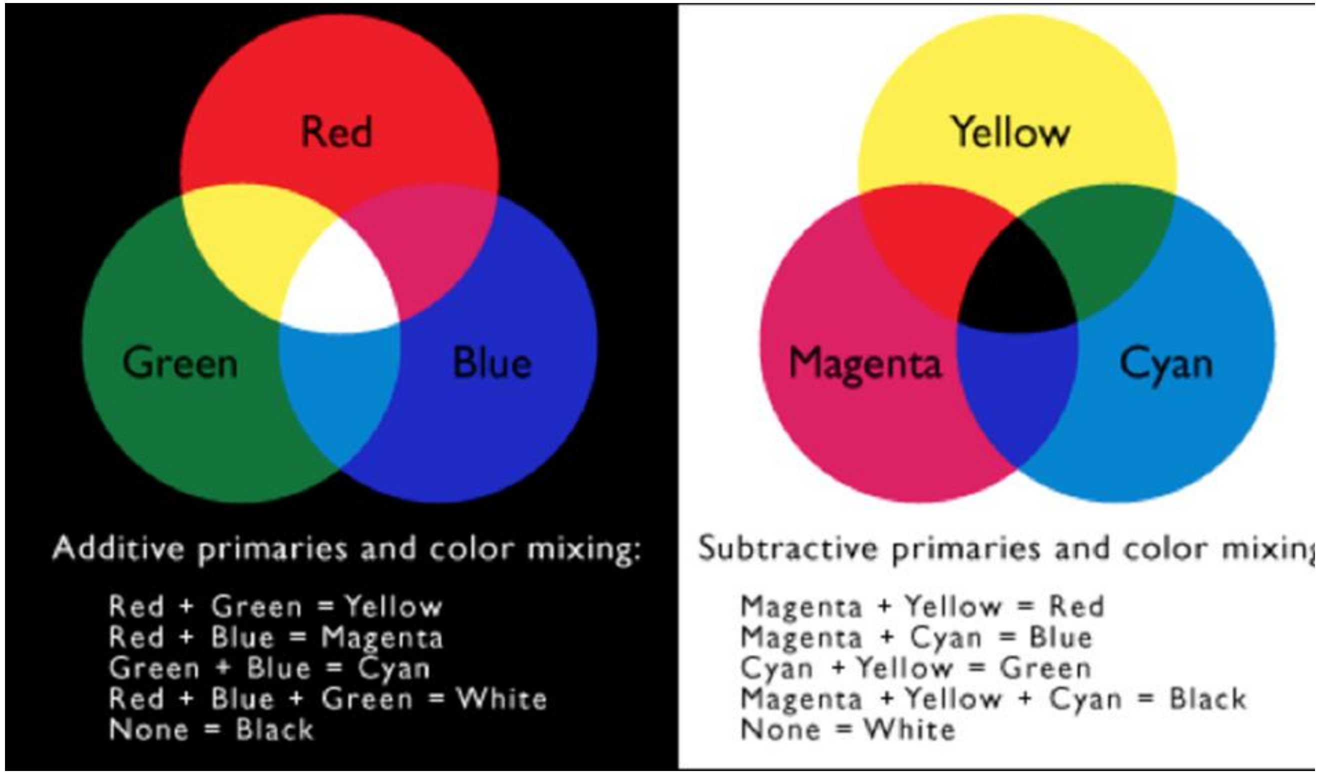
- To create documents with styles and formats, formatting codes must be embedded in the text.
    - Microsoft Word (DOCX)
    - Apple Pages (PAGES)
    - Adobe Acrobat (PDF)
    - Etc.




```
xmlns:wps="http://s fice/word/2010/wordprocessingShape" mc:Ignorable="w14
w15 wp14"><w:bod 49" w:rsidRPr="003E2449" w:rsidRDefault="003
E2449"><w:pPr><w:rPr><w:b/><w:sz w:val="32"/><w:szCs w:val="32"/></w:rPr></w:pPr><w:r
w:rsidRPr="003E2449"><w:rPr><w:b/><w:sz w:val="32"/><w:szCs w:val="32"/></w:rPr><w:t
xml:space="preserve">Roller Coasters </w:t></w:r></w:p><w:p w:rsidR="002107BB"
w:rsidRDefault="003E2449"><w:r w:rsidRPr="003E2449"><w:rPr><w:i/></w:rPr><w:t>Who wants to
save an old roller coaster?</w:t></w:r><w:r w:rsidRPr="003E2449"><w:t xml:space="preserve"> Leap-
the-Dips is the world's oldest roller coaster and, according to a spokesperson for the Leap-the-Dips
Preservation Foundation, one of the most historically significant. Built in 1902, Leap-the-Dips </w:t></
w:r><w:proofErr w:type="gramStart"/><w:r w:rsidRPr="003E2449"><w:t>is</w:t></w:r><w:proofErr
w:type="gramEnd"/><w:r w:rsidRPr="003E2449"><w:t xml:space="preserve"> "the sole survivor of a
style and technology that was represented in more than 250 parks in North America alone in the early
years of the amusement industry.</w:t></w:r><w:r w:rsidR="00881463"><w:t>"
```

Representing Images and Graphics

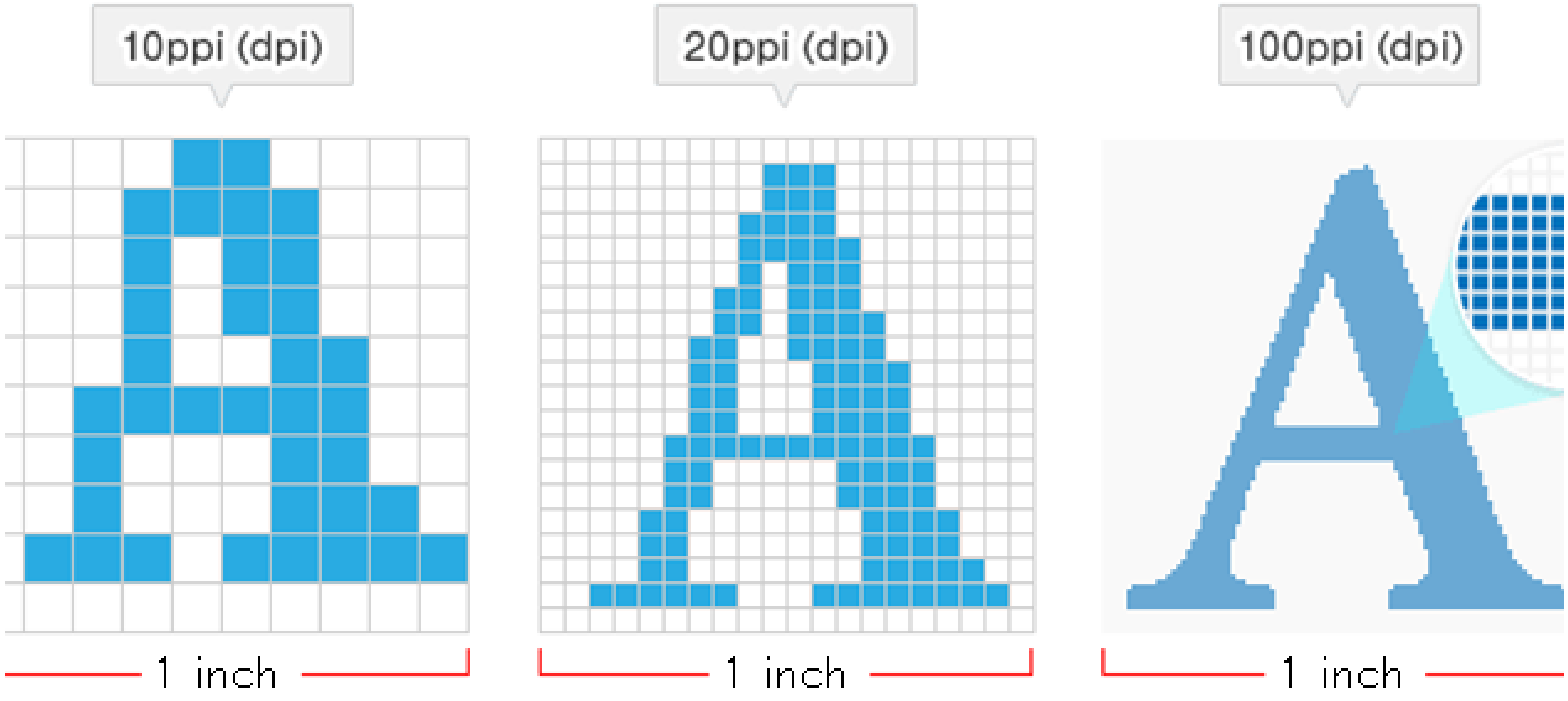
Color



- Color is expressed as a combination of three colors of light
- Each color value is represented as one byte, so each color value ranges from 0-255
- Two Types:
  - CMY (cyan-magenta-yellow) → Printers
  - RGB (red-green-blue) → Screen

Digitized Images and Graphics

- Digitizing a picture – Representing it as a collection of individual dots called pixels
- Resolution – The number of pixels used to represent a picture
- Raster Graphics – Storage of data on a pixel-by-pixel basis
    - Bitmap (BMP) → Contains the pixel color values of the images from left to right and from top to bottom
    - GIF → Each image is made up of only 256 colors
    - JPEG → Averages color hues over short distances
    - PNG → Like GIF but achieves greater compression with wider range of color depths



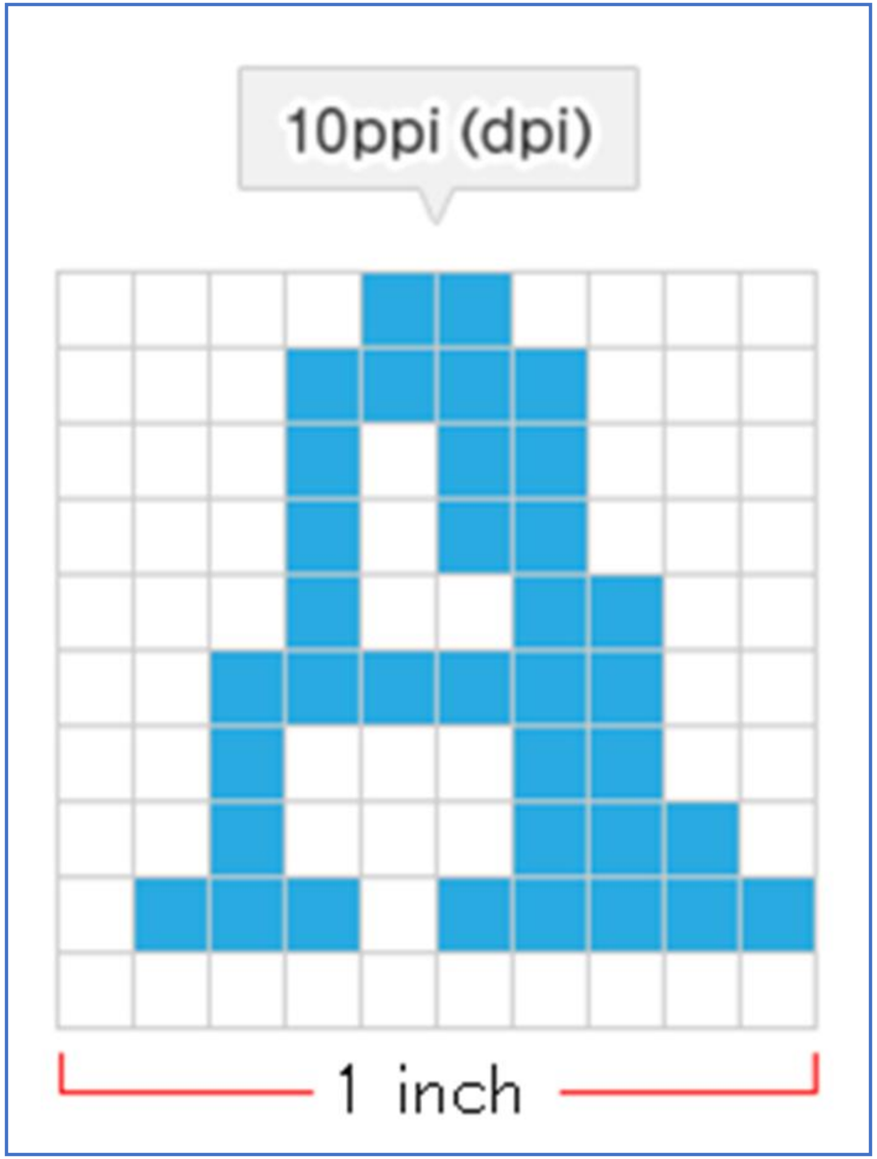
10ppi (dpi)
20ppi (dpi)
100ppi (dpi)
1 inch
1 inch
1 inch

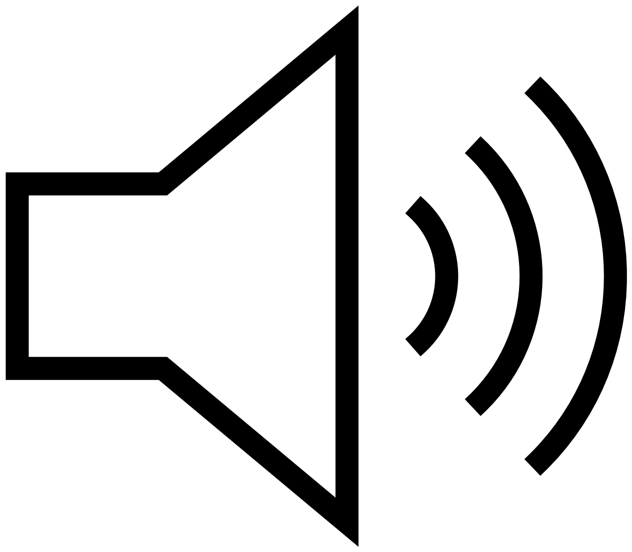
Storage

- Image Specifications:
  - Resolution: 10ppi (10p x 10p)
  - RGB: 3 bytes

How many bytes would be needed to store this image?

- 10p x 10p = 100 pixels
- Each pixel has 3 bytes → 300 bytes

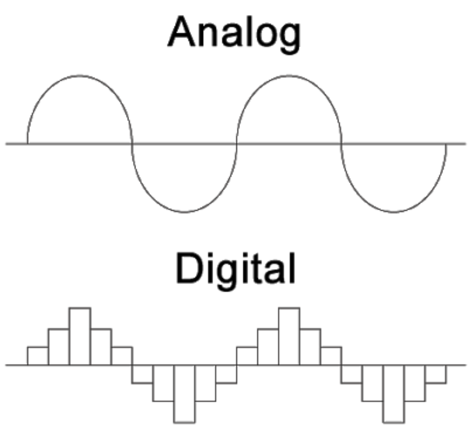


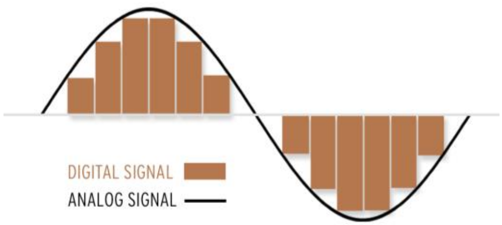


Representing Sound

Analog vs Digital

- Analog
  - Infinite
  - Real-world
- Digital
  - Finite (discrete)





Sound

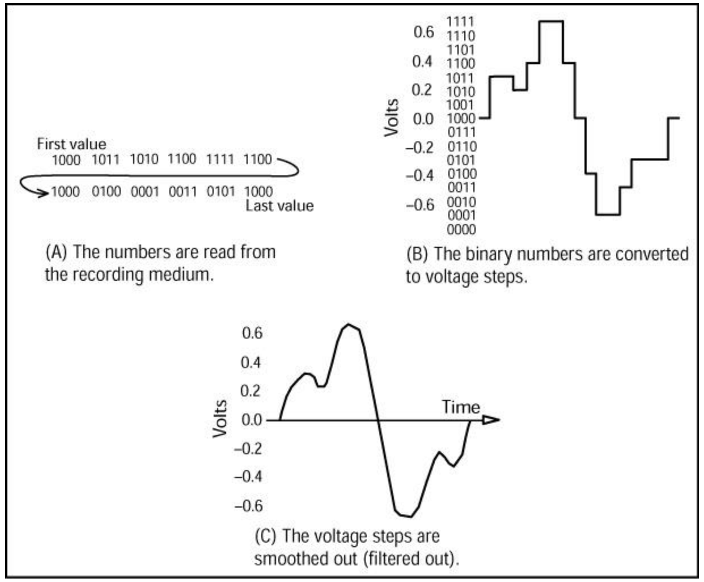
We perceive sound when a series of air compressions vibrate a membrane in our ear, which sends signals to our brain. (Analog)

How do computers perceive sound?

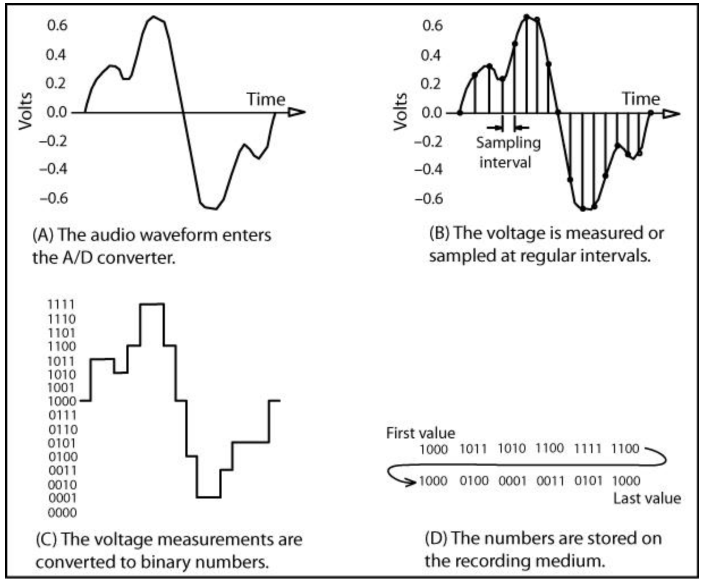
Example: A microphone converts the sound pressure and vibrations into an electrical signal.

Analog ↔ Digital

- Analog-to-Digital Convertor (ADC)
- Digitize the signal by sampling
    - Periodically measure the voltage
    - Record the numeric value
- Digital-to-Analog Convertor (DAC)
    - Convert back to voltage



(A) The numbers are read from the recording medium.
(B) The binary numbers are converted to voltage steps.
(C) The voltage steps are smoothed out (filtered out).

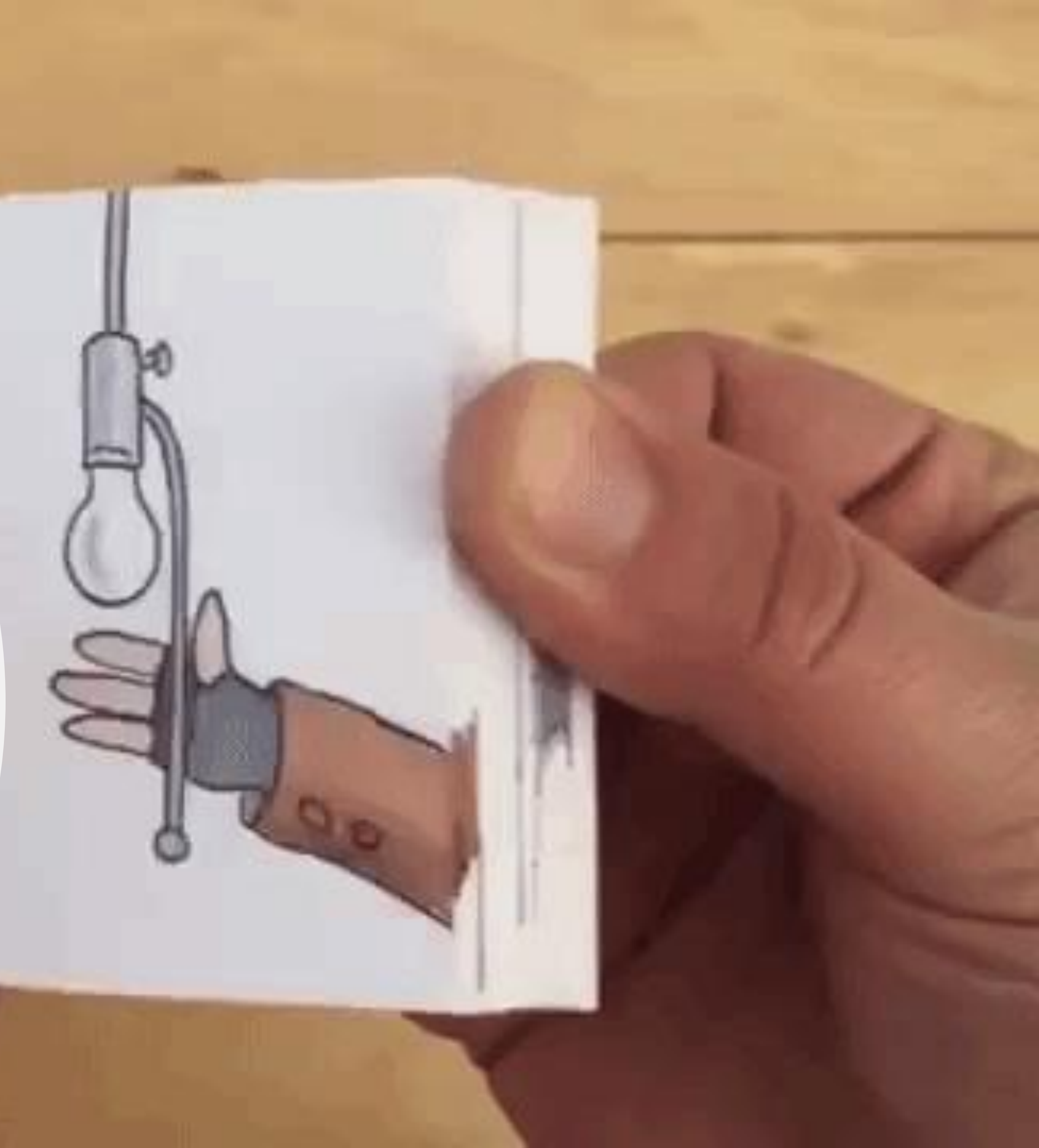


(A) The audio waveform enters the A/D converter.
(B) The voltage is measured or sampled at regular intervals.
(C) The voltage measurements are converted to binary numbers.
(D) The numbers are stored on the recording medium.

Representing Video

What Is A Video?

- Basically a series of pictures being “flipped” through with audio
- The same/similar concepts for image and audio representation apply to video



Digital Video

- Made up of individual frames (images), each one representing a time slice of the scene with audio stored as a separate stream
- Frame Rate: frequency at which frames are displayed
- Video Quality:
    - The higher the frame rate the smoother the video will appear
    - The higher resolution the higher the quality of the image